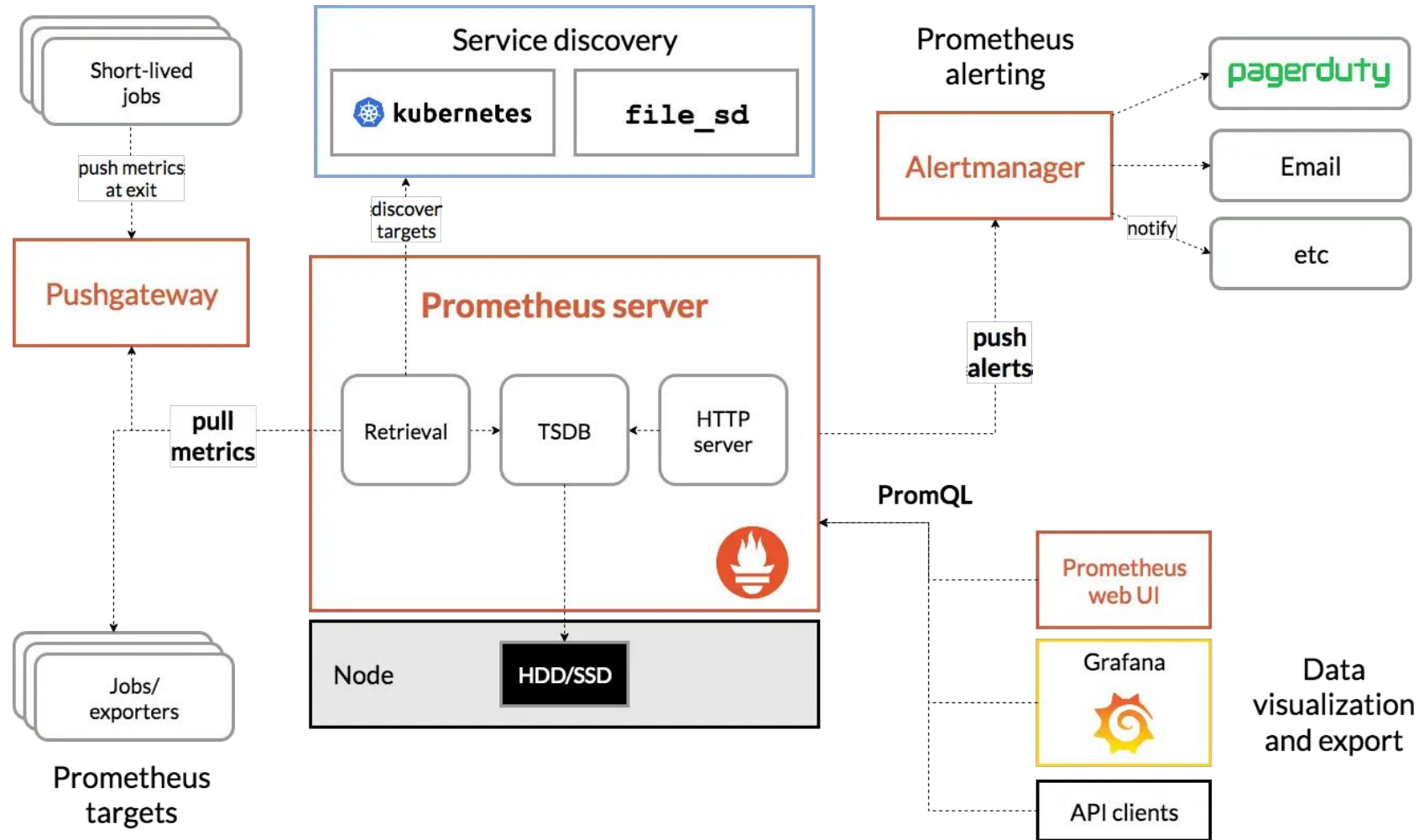




Prometheus Architecture with Grafana



Installing Prometheus with Docker

```
$ docker run -d -p 9090:9090 -v /path/to/prometheus.yml:/etc/prometheus/prometheus.yml prom/prometheus
```

```
$ vim prometheus.yml
```

```
global:
  scrape_interval: 15s
  external_labels:
    monitor: 'codelab-monitor'

scrape_configs:
  - job_name: 'prometheus'
    scrape_interval: 5s
    static_configs:
      - targets: ['10.148.0.5:9100', '10.148.0.2:9100']
```

Add Basic Authentication User to Prometheus

- `vim /home/kao.leangseng/monitoring/prom/web.yml`
`basic_auth_users:`
`dara: $2a$12$YxoDjQWTaPyP9lyyHoRyMOV3Zr/pUxnI6r56Grt24N.SsGrbyBM7G`
- Generate Bcrypt hash online: <https://bcrypt-generator.com/>
- `docker run -d -p 9090:9090 -v /home/kao.leangseng/monitoring/prometheus:/etc/prometheus -v /home/kao.leangseng/monitoring/prom/prometheus.yml:/etc/prometheus/prometheus.yml -v /home/kao.leangseng/monitoring/prom/web.yml:/etc/prometheus/web.yml --name prometheus prom/prometheus --config.file=/etc/prometheus/prometheus.yml --web.config.file=/etc/prometheus/web.yml --web.enable-lifecycle`

Installing Node Exporter in all required nodes

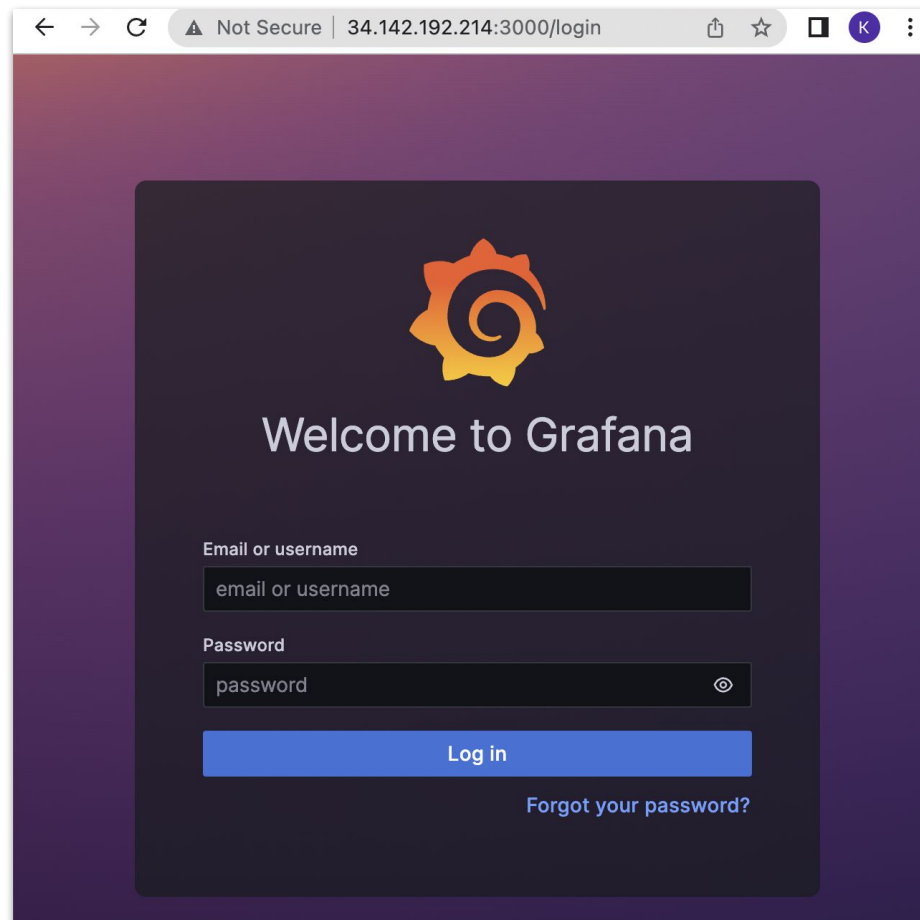
- **Visit:** https://prometheus.io/download/#node_exporter
- wget
https://github.com/prometheus/node_exporter/releases/download/v1.6.0/node_exporter-1.6.0.linux-amd64.tar.gz
- cd node_exporter-1.6.0.linux-amd64
- tar -xvf [node_exporter-1.6.0.linux-amd64.tar.gz](https://github.com/prometheus/node_exporter/releases/download/v1.6.0/node_exporter-1.6.0.linux-amd64.tar.gz)
- ./node-exporter &

OR: <https://ourcodeworld.com/articles/read/1686/how-to-install-prometheus-node-exporter-on-ubuntu-2004>

Installing Grafana with Docker

```
$ docker run -d -p 3000:3000 --name=grafana --user "$(id -u)" --volume  
"/home/kao.leangseng/monitoring/grafana:/var/lib/grafana" grafana/grafana-enterprise
```

- Username and Password: **admin**

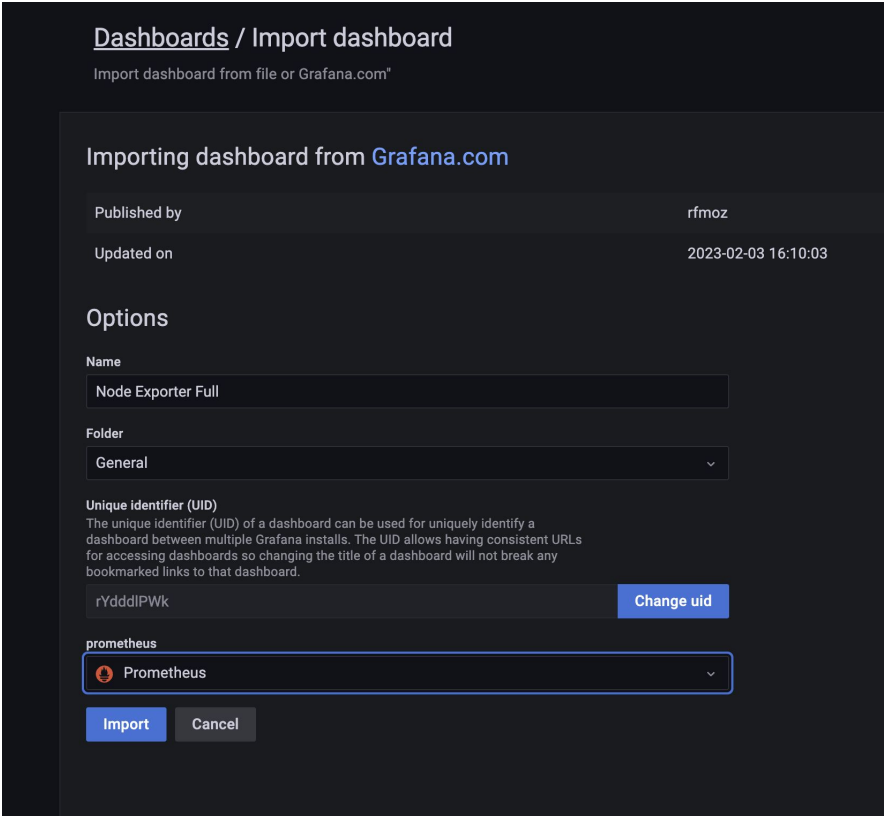
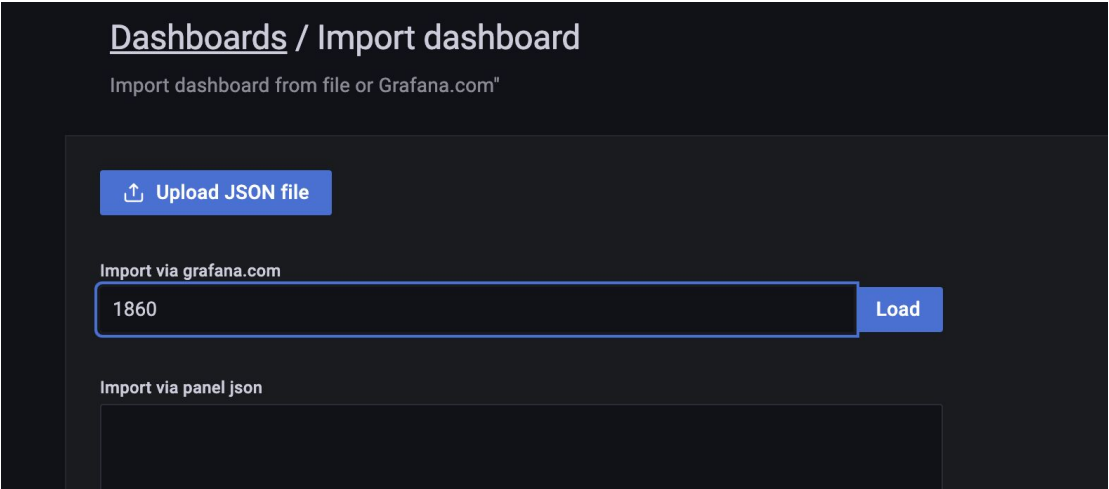
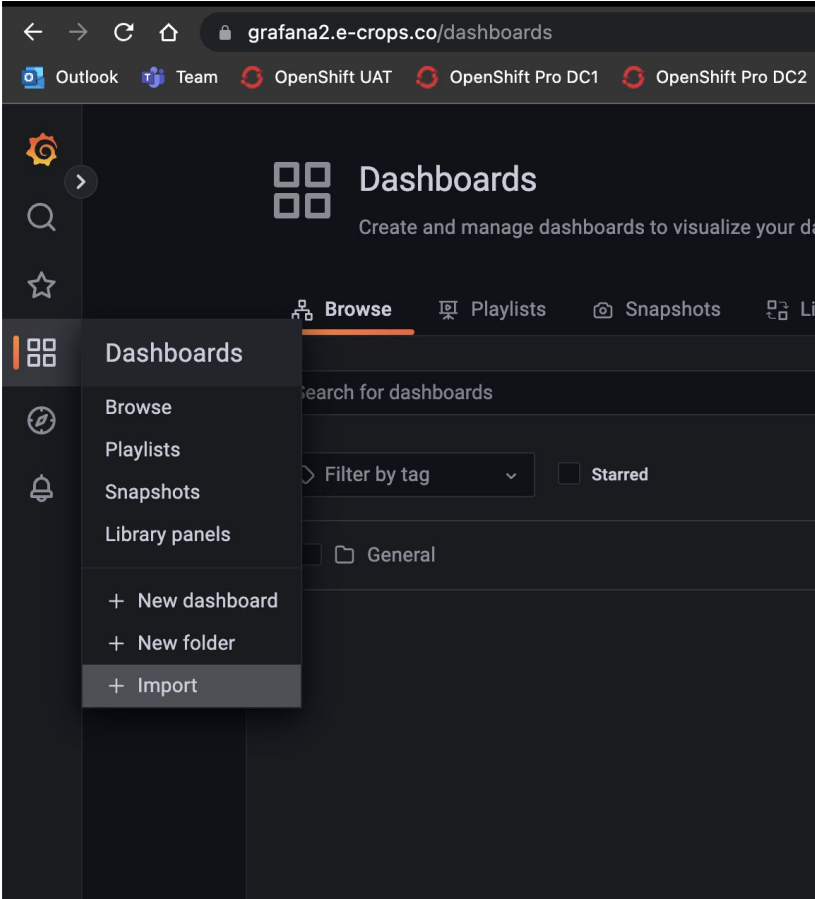


Grafana Dashboard Addons

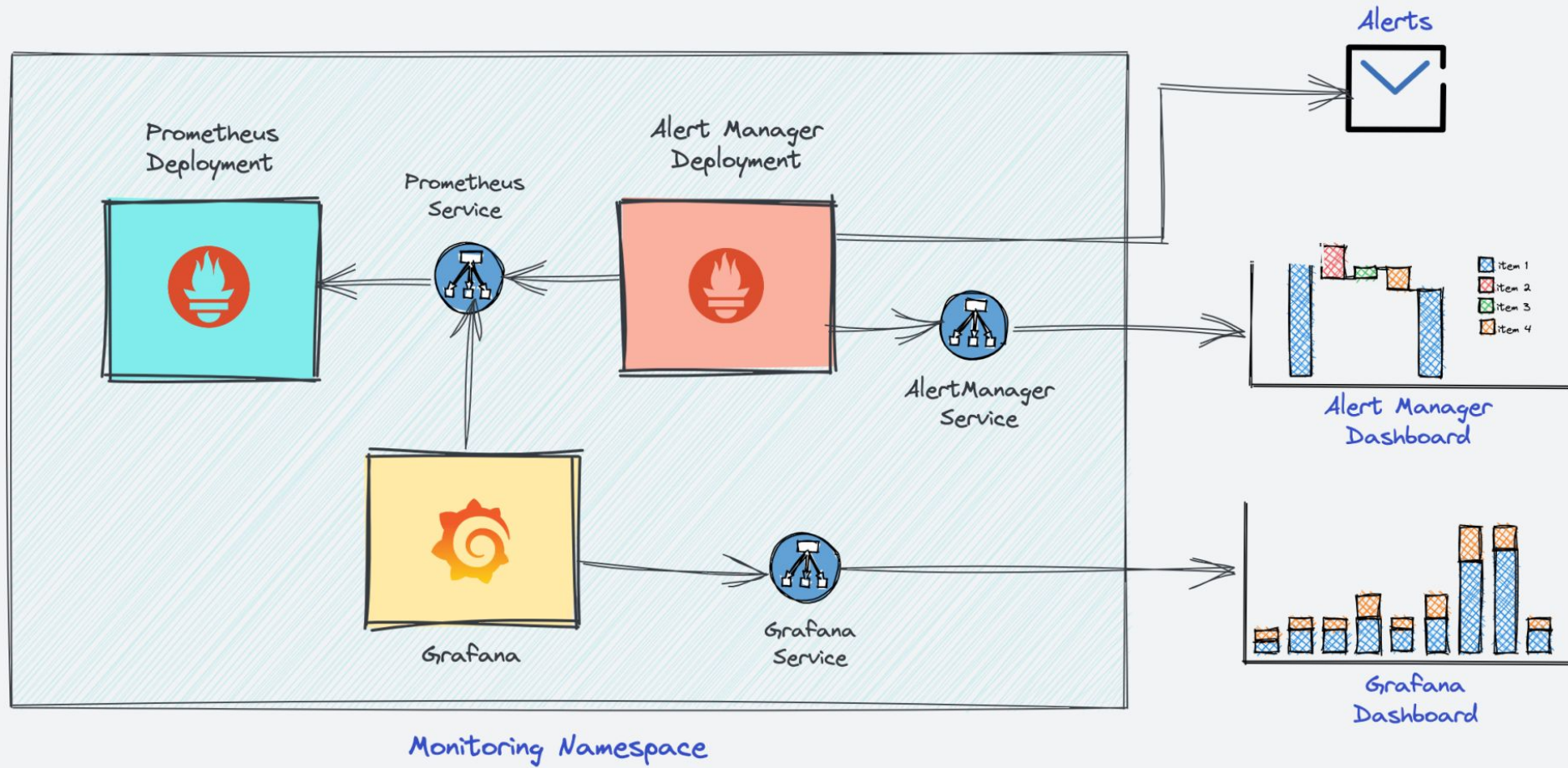
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7

Grafana Dashboard Addons



Using Kube-Prometheus-Stack



Using Kube-Prometheus-Stack

The [kube-prometheus-stack](#) Helm chart is the simplest way to bring up a complete Prometheus stack inside your Kubernetes cluster. It bundles several different components in one automated deployment:

- **Prometheus** – Prometheus is the time series database that scrapes, stores, and exposes the metrics from your Kubernetes environment and its applications.
- **Node-Exporter** – Prometheus works by scraping data from a variety of configurable sources called exporters. [Node-Exporter](#) is an exporter which collects resource utilization data from the Nodes in your Kubernetes cluster. The [kube-prometheus-stack](#) chart automatically deploys this exporter and configures your Prometheus instance to scrape it.
- **Kube-State-Metrics** – [Kube-State-Metrics](#) is another exporter that supplies data to Prometheus. It exposes information about the API objects in your Kubernetes cluster, such as Pods and containers.
- **Grafana** – Although you can directly query Prometheus, this is often tedious and repetitive. [Grafana](#) is an observability platform that works with several data sources, including Prometheus databases. You can use it to create dashboards that surface your Prometheus data.
- **Alertmanager** – [Alertmanager](#) is a standalone Prometheus component that provides notifications when metrics change. You can use it to get an email when CPU utilization spikes or a Slack notification if a Pod is evicted.

Installing Kube-Prometheus-Stack

```
$ helm repo add prometheus-community https://prometheus-community.github.io/helm-charts
```

```
$ helm repo update
```

```
$ helm install kube-prometheus-stack --create-namespace --namespace kube-prometheus-stack  
prometheus-community/kube-prometheus-stack
```

```
$ kubectl -n kube-prometheus-stack get pods
```

```
$ kubectl -n kube-prometheus-stack get pods
```

NAME	READY	STATUS	RESTARTS	AGE
alertmanager-kube-prometheus-stack-alertmanager-0	2/2	Running	1 (66s ago)	83s
kube-prometheus-stack-grafana-5cd658f9b4-cln2c	3/3	Running	0	99s
kube-prometheus-stack-kube-state-metrics-b64cf5876-52j8l	1/1	Running	0	99s
kube-prometheus-stack-operator-754ff78899-669k6	1/1	Running	0	99s
kube-prometheus-stack-prometheus-node-exporter-vdgrg	1/1	Running	0	99s
prometheus-kube-prometheus-stack-prometheus-0	2/2	Running	0	83s

Installing Kube-Prometheus-Stack

```
apiVersion: cert-manager.io/v1
kind: ClusterIssuer
metadata:
  name: prometheus-clusterissuer
  namespace: kube-prometheus-stack
spec:
  acme:
    server: https://acme-v02.api.letsencrypt.org/directory
    email: leangsengk90@gmail.com
    privateKeySecretRef:
      name: prometheus-key
    solvers:
      - http01:
          ingress:
            class: nginx
```

Installing Kube-Prometheus-Stack

```
apiVersion: cert-manager.io/v1
kind: Certificate
metadata:
  name: prometheus-certificate
  namespace: kube-prometheus-stack
  annotations:
    cert-manager.io/cluster-issuer: prometheus-clusterissuer
spec:
  dnsNames:
    - prometheus.e-crops.co
    - grafana2.e-crops.co
  secretName: prometheus-tls
  issuerRef:
    name: prometheus-clusterissuer
    kind: ClusterIssuer
```

Installing Kube-Prometheus-Stack

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: prometheus-ingress
  namespace: kube-prometheus-stack
spec:
  tls:
    - hosts:
        - prometheus.e-crops.co
      secretName: prometheus-tls
  rules:
    - host: prometheus.e-crops.co
      http:
        paths:
          - pathType: Prefix
            path: "/"
            backend:
              service:
                name: kube-prometheus-stack-prometheus
                port:
                  number: 9090
    - host: grafana2.e-crops.co
      http:
        paths:
          - pathType: Prefix
            path: "/"
            backend:
              service:
                name: kube-prometheus-stack-grafana
                port:
                  number: 80
```

Prometheus

Go to: <https://prometheus.e-crops.co>



Prometheus

Go to: <https://prometheus.e-crops.co/targets?search=>

Prometheus Alerts Graph Status ▾ Help Classic UI

Targets

All Unhealthy Expand All

kube-state-metrics (0/1 up) show more

kubernetes-apiservers (2/2 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
https://10.0.1.161/metrics	UP	instance="10.0.1.161:443" job="kubernetes-apiservers"	15.760s ago	143.912ms	
https://10.0.2.10/metrics	UP	instance="10.0.2.10:443" job="kubernetes-apiservers"	15.116s ago	149.923ms	

kubernetes-cadvisor (3/3 up) show less

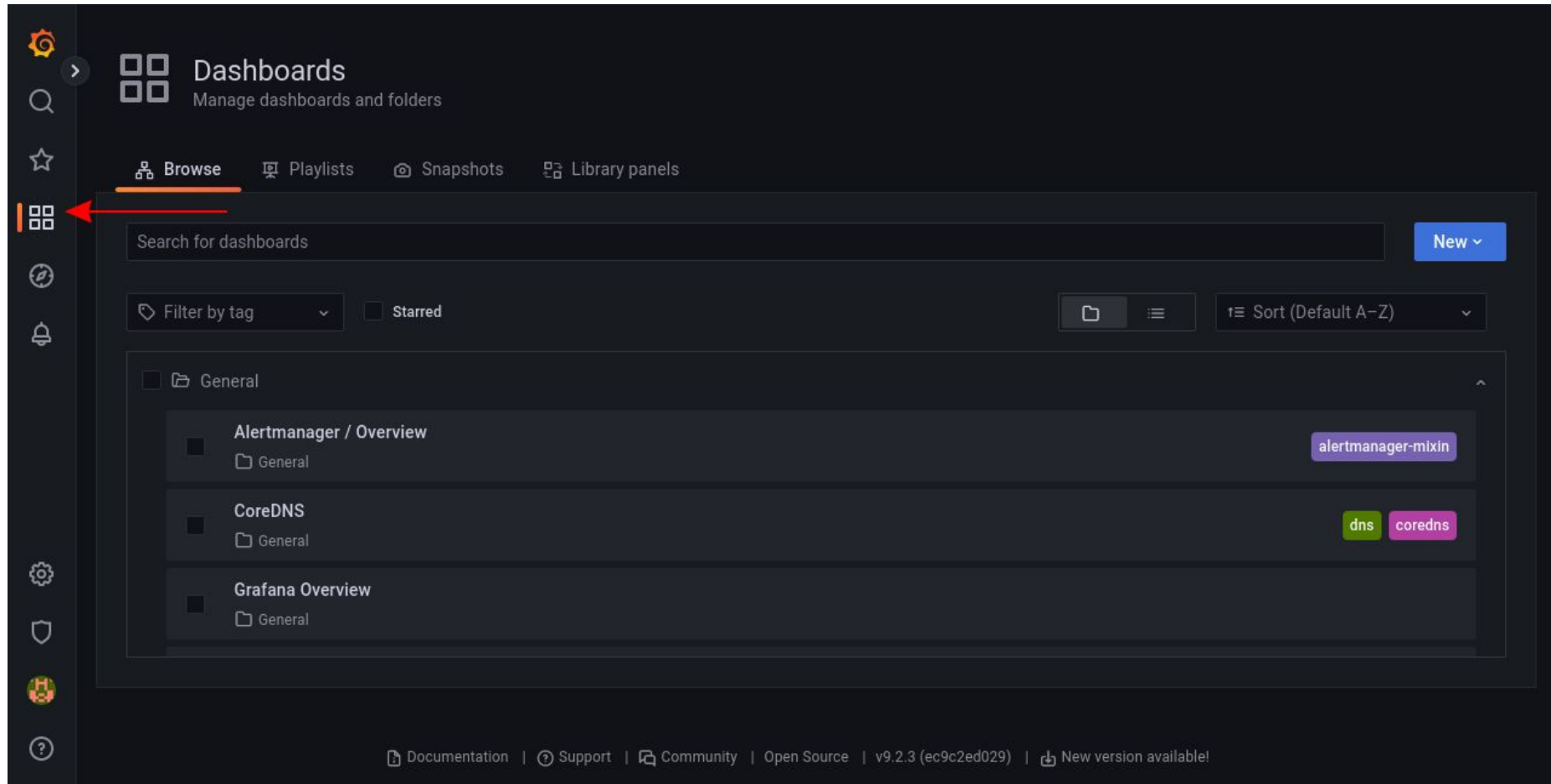
Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
https://kubernetes.default.svc/api/v1/nodes/ip-10-0-1-48.us-west-2.compute.internal/proxy/metrics/cadvisor	UP	alpha_eksctl_io_cluster_name="eks-spot-cluster" alpha_eksctl_io_nodegroup_name="ng-spot-02" beta_kubernetes_io_arch="amd64" beta_kubernetes_io_instance_type="t3.large" beta_kubernetes_io_os="linux" eks_amazonaws_com_capacityType="SPOT"	14.267s ago	138.514ms	

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16

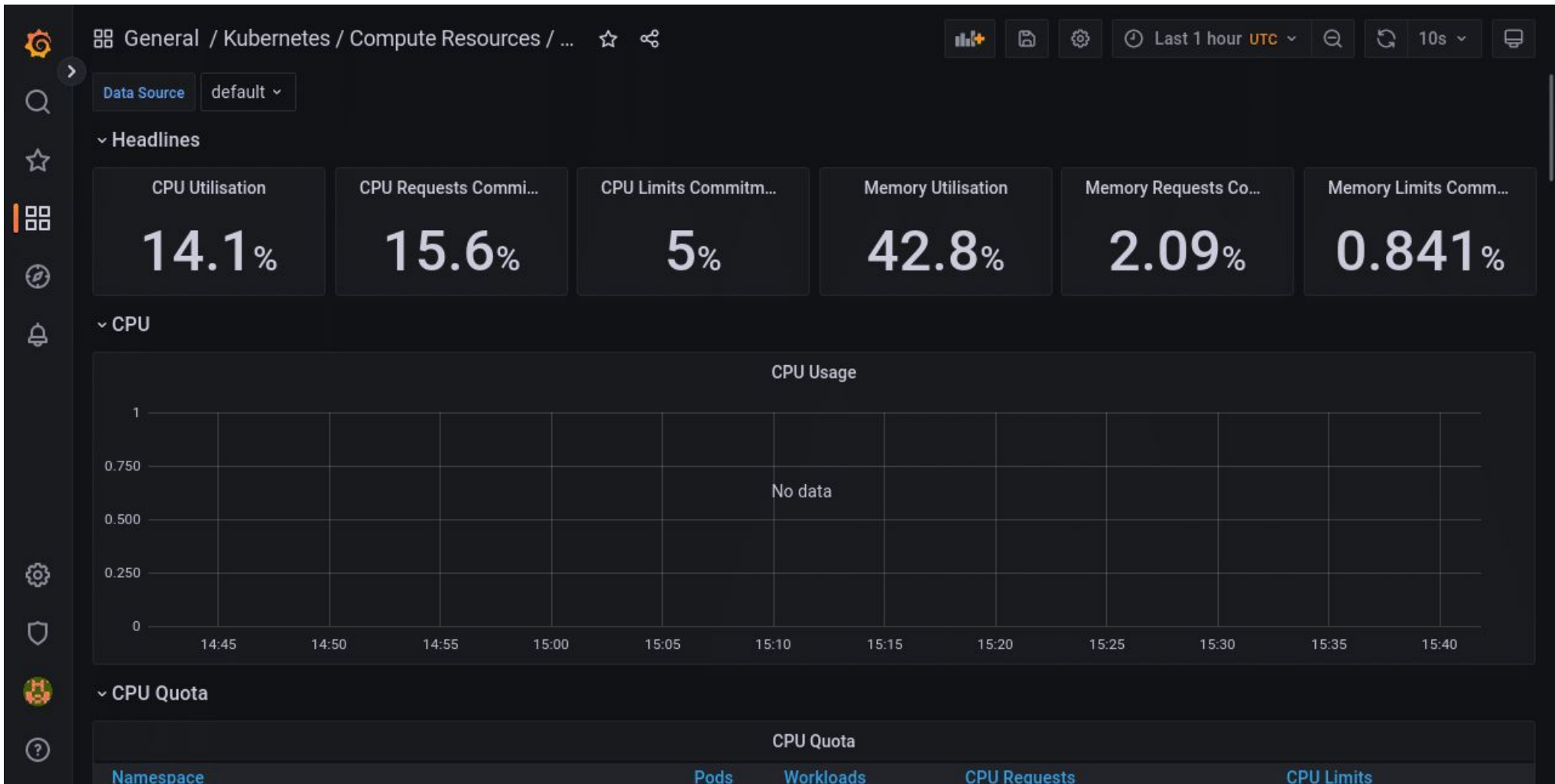
Grafana

The default user account is **admin** with a password is **prom-operator**.



Grafana

View any existing built-in dashboard



Grafana (muting the notification)

Silences

Stop notifications from one or more alerting rules

Search by matchers ⓘ

Choose Alertmanager

Alertmanager

+ Add Silence

State	Matching labels	Alerts	Schedule	Actions
> Active	severity=critical	3	2024-05-30 10:18 -2024-05-30 12:12	🔔 Unsilence

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with a white atomic symbol. The word 'ISTAD' is written in red across the bottom of the inner circle. The text 'Thank you' is overlaid in the center in a bold, dark blue font.

Thank you

Begin with the theory, Start with the practical